

Solar & Battery Energy Storage System (BESS) Dashboard – KPI Summary

Project Overview

This dashboard analyzes the performance of a decentralized Solar and Battery Energy Storage System (BESS) project implemented across commercial and institutional facilities in Kenya. The purpose of the dashboard is to evaluate technical efficiency, financial impact, and environmental sustainability.

Selected KPIs

1. Energy Generated (kWh)

This KPI measures the total amount of solar energy generated by the system over time. It helps management monitor operational efficiency and evaluate overall energy production performance.

2. Diesel Savings

This KPI tracks the reduction in diesel fuel usage and associated cost savings achieved through solar and battery deployment. It helps assess the financial benefits and operational cost optimization of the project.

3. Battery Utilization (%)

Battery utilization indicates how efficiently the Battery Energy Storage System is being used. Higher utilization reflects improved energy storage optimization and better backup power management.

4. CO₂ Emissions Avoided (kg)

This KPI measures the reduction in carbon emissions resulting from reduced diesel generator dependence. It helps evaluate the environmental sustainability and clean energy impact of the

project.

Dashboard Insights

- Solar energy generation showed a steady increase throughout the year.
- Diesel dependency reduced significantly over time, resulting in cost savings.
- Battery utilization remained consistently high, indicating efficient storage performance.
- CO₂ emission reduction increased steadily, supporting sustainability goals.

Tools Used

- Tableau Public
- Mock dataset generated for analytical visualization
- CSV dataset

Conclusion

The dashboard provides a clear overview of operational, financial, and environmental performance for the Solar & BESS project. These insights can support management decision-making, investment evaluation, and long-term scalability planning.